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(54) **MATTRESS ASSEMBLY**

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(60) Provisional application No. 62/580,067, filed on Nov. 1, 2017.

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A47C 27/06 (2006.01)
A47C 27/12 (2006.01)
A47C 27/15 (2006.01)

(52) **U.S. Cl.**
CPC *A47C 27/001* (2013.01); *A47C 27/06* (2013.01); *A47C 27/121* (2013.01); *A47C 27/122* (2013.01); *A47C 27/15* (2013.01)

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A47C 27/063; A47C 27/121; A47C 27/122; A47C 27/15; A47C 27/061; A47C 27/062; A47C 27/064; A47C 27/065; A47C 27/066

See application file for complete search history.

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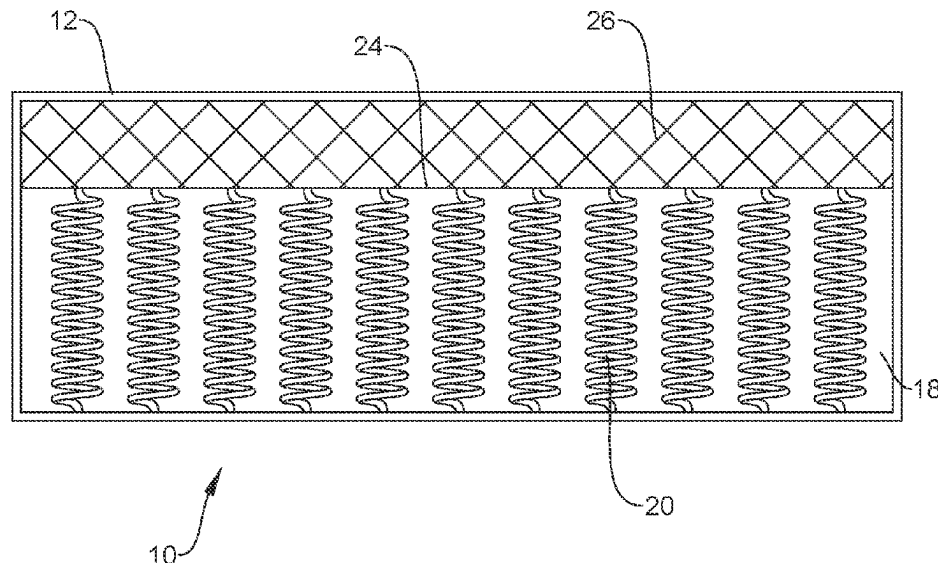
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(57) **ABSTRACT**

A mattress assembly includes a cover, a spring pack positioned within the cover, and an inner core positioned within the cover. The inner core includes opposite first and second portions. The first portion has a firmness that is different than a firmness of the second portion. The inner core is configured to be moved between a first configuration in which the first portion directly engages the spring pack and a second configuration in which the second portion directly engages the spring pack to alter a firmness of a sleep surface of the mattress. Methods of use are disclosed.

20 Claims, 7 Drawing Sheets



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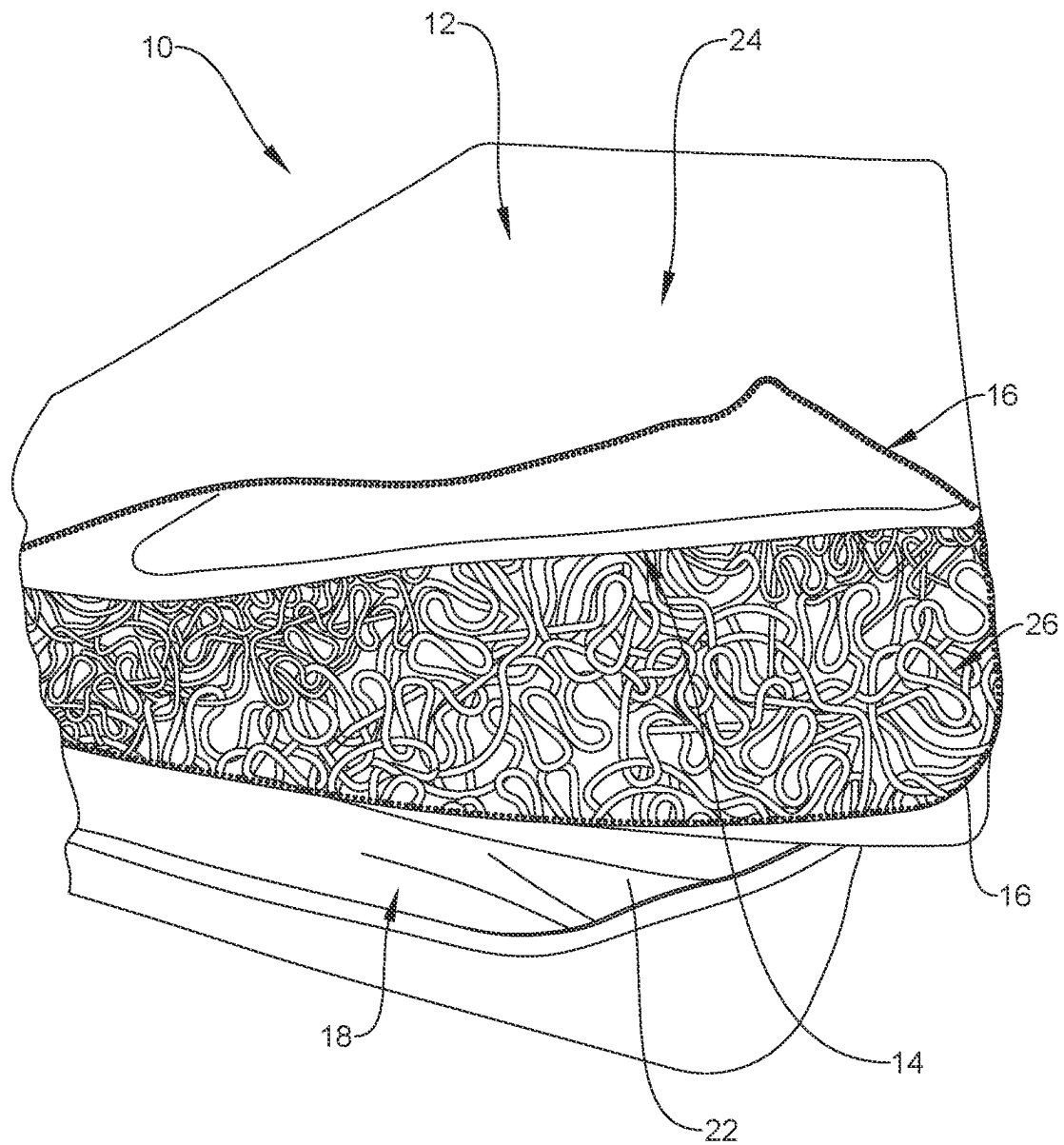


FIG. 1

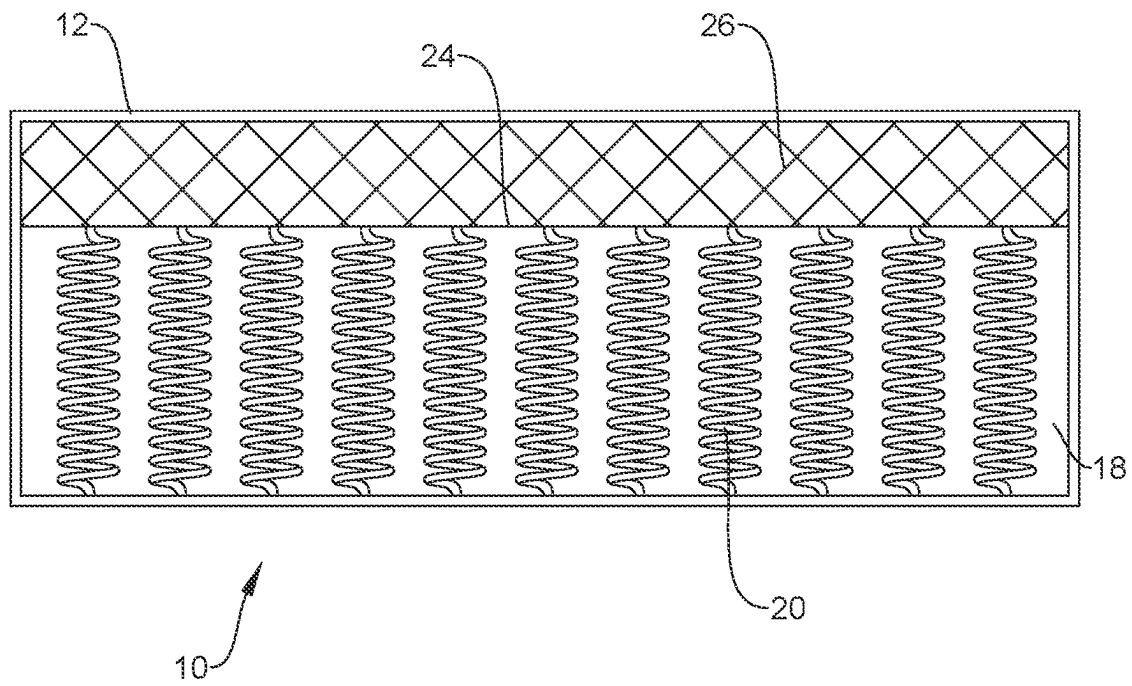


FIG. 2

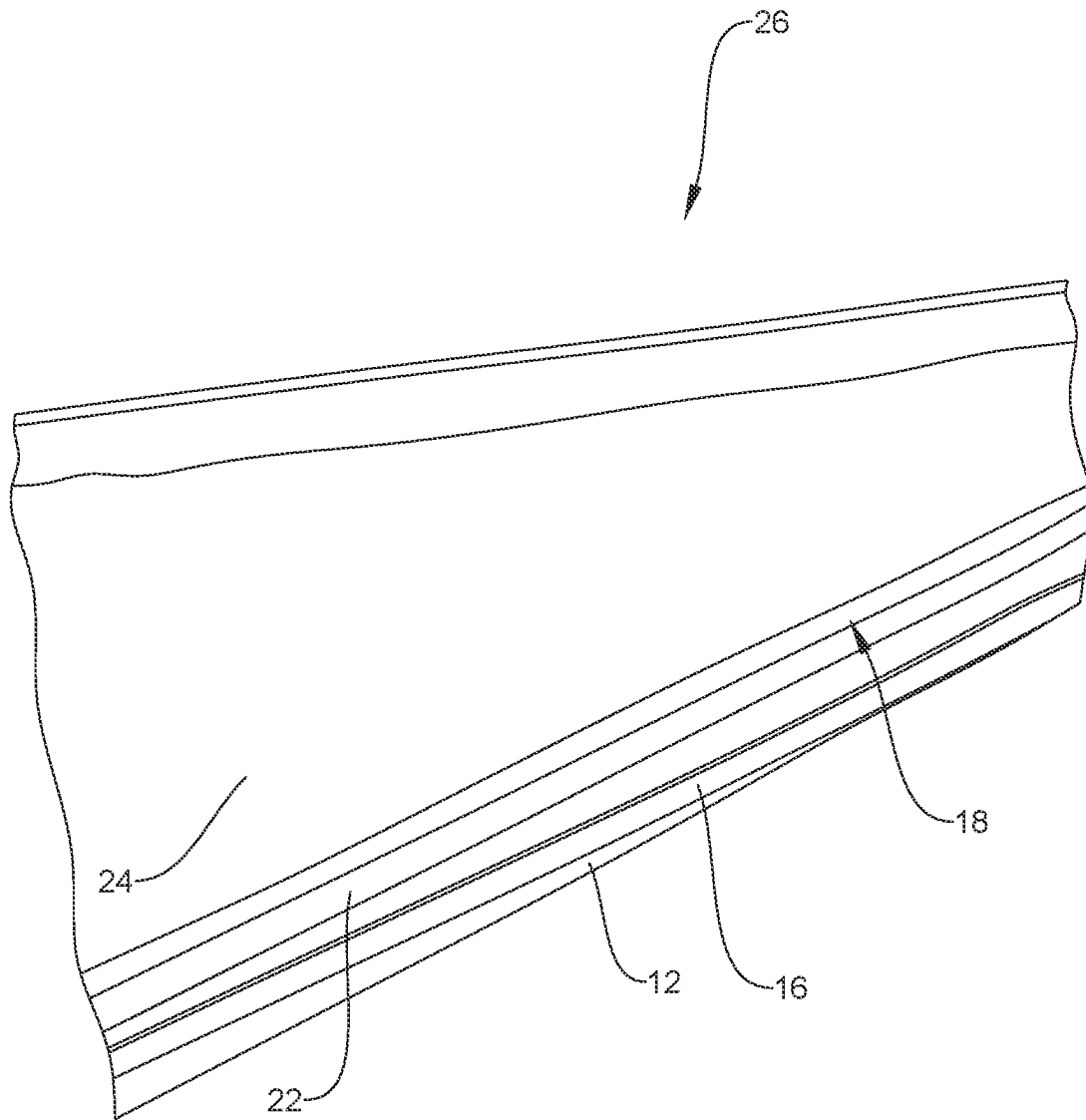


FIG. 3

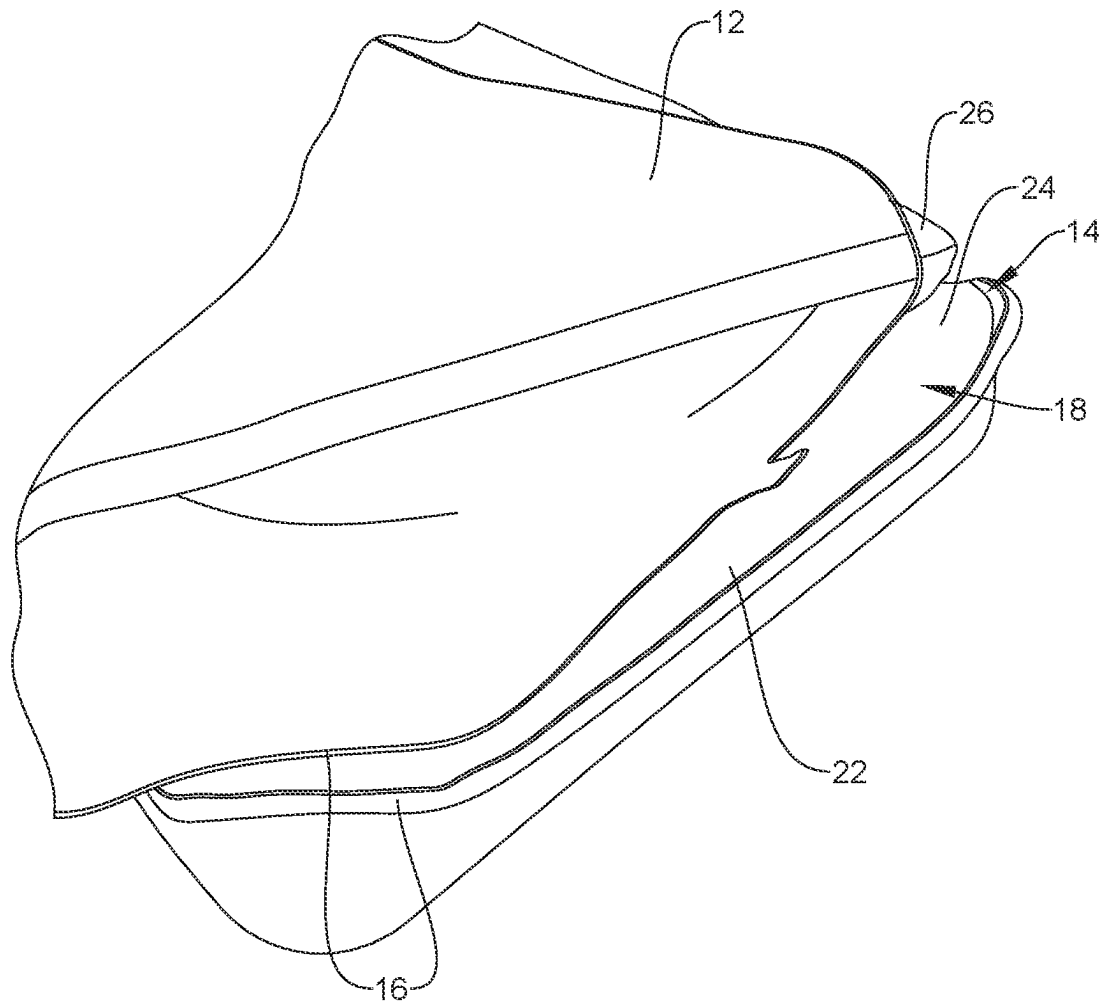


FIG. 4

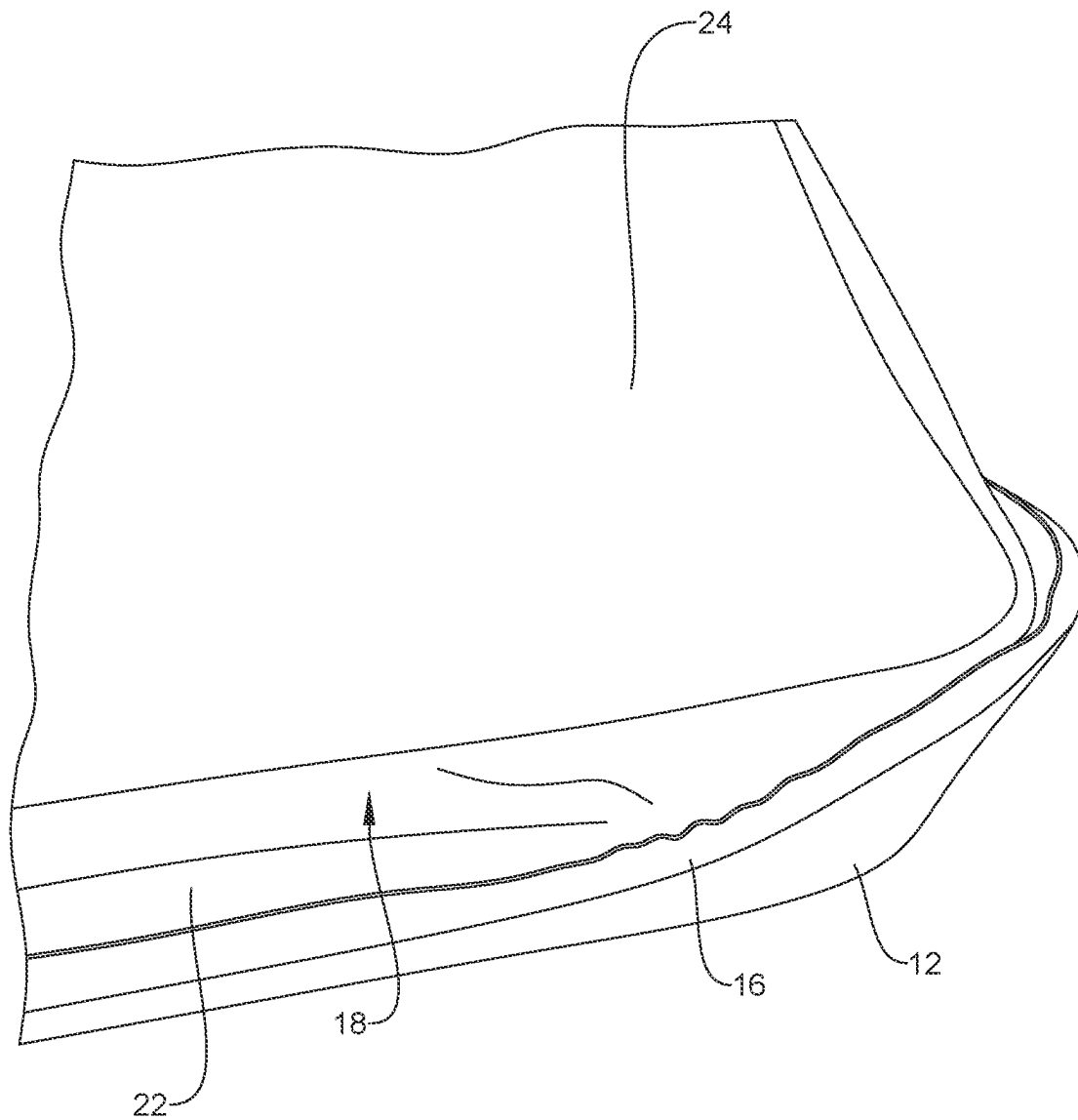


FIG. 5

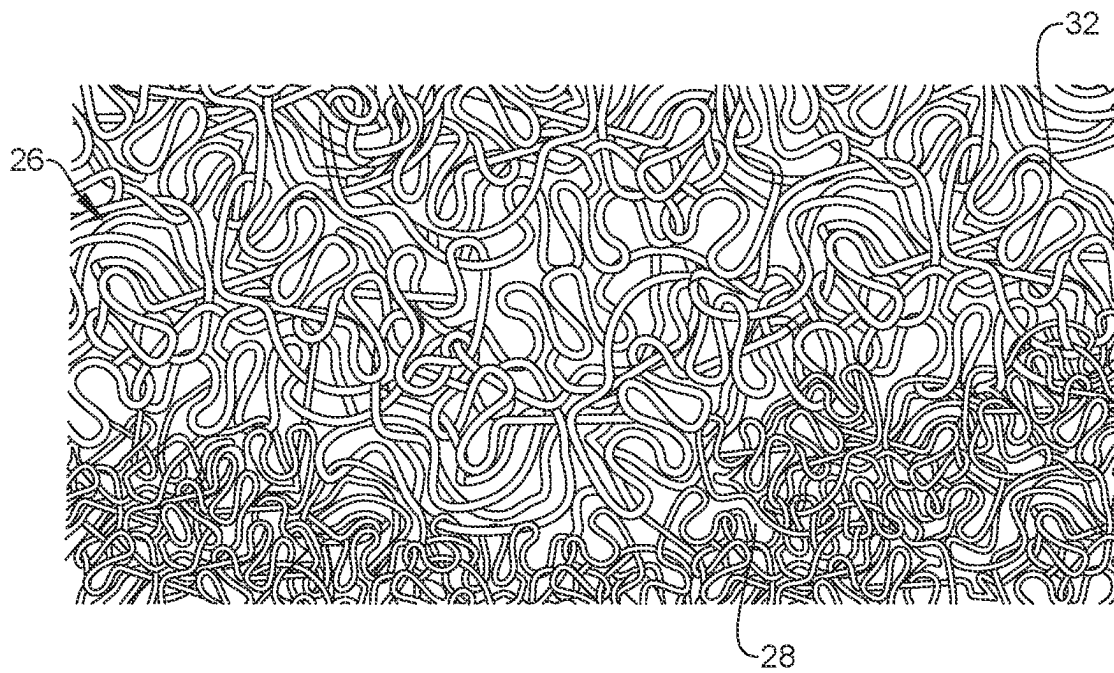


FIG. 6

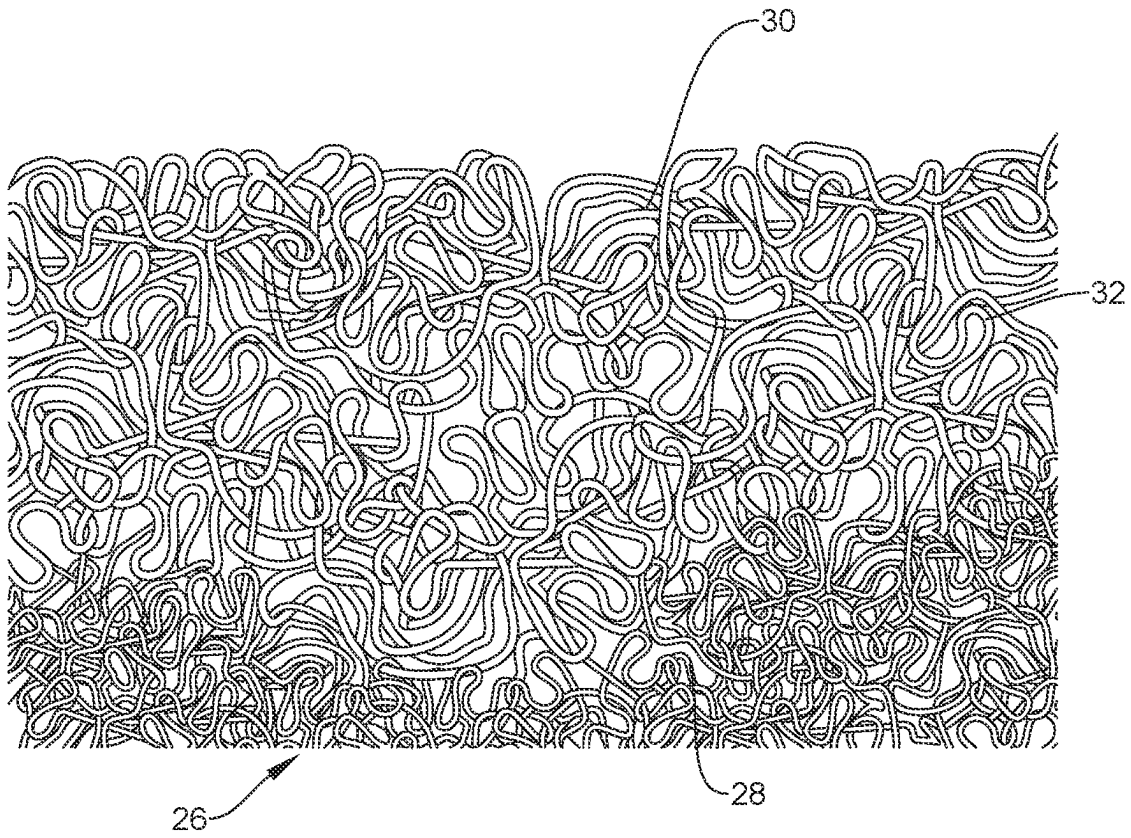


FIG. 7

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MATTRESS ASSEMBLY**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a continuation application of U.S. patent application Ser. No. 16/177,949, filed on Nov. 1, 2018, which is expressly incorporated by reference herein, in its entirety.

TECHNICAL FIELD

The present disclosure generally relates to mattresses, and more particularly to mattress assemblies that include a dual-side mattress having two different comfort choices.

BACKGROUND

Sleep is critical for people to feel and perform their best, in every aspect of their lives. Sleep is an essential path to better health and reaching personal goals. Indeed, sleep affects everything from the ability to commit new information to memory to weight gain. It is therefore essential for people to use bedding that suit both their personal sleep preference and body type in order to achieve comfortable, restful sleep.

Infants and toddlers often require firmer mattresses than older children. In order to prevent having to purchase a new mattress when children become older, some mattresses include a first side having a firmness that is suited for toddlers and a second side having a firmness that is suited for older children. In that young children are likely to wet the bed, such mattresses may include a waterproof cover to prevent urine from getting into the mattress. However, such waterproof covers prevent body heat from ventilating through the mattress and are hence uncomfortable to sleep on. Furthermore, conventional mattresses typically include a cover that is sewn onto an inner core of the mattress, thus requiring that the cover be flipped when the mattress is flipped. This disclosure describes an improvement over these prior art technologies.

SUMMARY

In one embodiment, in accordance with the principles of the present disclosure, a mattress assembly is provided that includes a cover, a spring pack positioned within the cover, and an inner core positioned within the cover. The inner core includes opposite first and second portions. The first portion has a firmness that is different than a firmness of the second portion. The inner core is configured to be moved between a first configuration in which the first portion directly engages the spring pack and a second configuration in which the second portion directly engages the spring pack to alter a firmness of a sleep surface of the mattress.

In one embodiment, in accordance with the principles of the present disclosure, a mattress assembly is provided that includes a cover, a spring pack positioned within the cover, and an inner core positioned within the cover. The cover comprises a porous, non-waterproof material and a zipper. The spring pack comprises opposite first and second surfaces and a plurality of springs positioned between the first and second surfaces. The first surface defines a waterproof barrier. The inner core comprises a matrix of polyurethane fibers. The matrix comprises opposite first and second portions. The fibers that make up the first portion are more densely packed than the fibers that make up the second

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portion such that the first portion has a firmness that is greater than a firmness of the second portion. The inner core is configured to be moved between a first configuration in which the first portion directly engages the waterproof barrier and a second configuration in which the second portion directly engages the waterproof barrier to alter a firmness of a sleep surface of the mattress.

In one embodiment, in accordance with the principles of the present disclosure, a mattress assembly is provided that includes a cover, a spring pack positioned within the cover, and an inner core positioned within the cover. The cover comprises a porous, non-waterproof material and a zipper. The spring pack comprises opposite first and second surfaces and a plurality of springs positioned between the first and second surfaces. The first surface defines a waterproof barrier. The inner core comprises a mat of electrospun polyurethane fibers. The mat comprises opposite first and second portions. The fibers that make up the first portion are more densely packed than the fibers that make up the second portion such that the first portion has a firmness that is greater than a firmness of the second portion. The inner core is configured to be moved between a first configuration in which the first portion directly engages the waterproof barrier and a second configuration in which the second portion directly engages the waterproof barrier to alter a firmness of a sleep surface of the mattress.

BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure will become more readily apparent from the specific description accompanied by the following drawings, in which:

FIG. 1 is a perspective view of one embodiment of a mattress assembly in accordance with the present principles of the present disclosure;

FIG. 2 is a cross sectional view of the mattress assembly shown in FIG. 1;

FIG. 3 is a perspective view of components of the mattress assembly shown in FIG. 1;

FIG. 4 is a perspective view of components of the mattress assembly shown in FIG. 1;

FIG. 5 is a perspective view of components of the mattress assembly shown in FIG. 1;

FIG. 6 is a perspective view of a component of the mattress assembly shown in FIG. 1; and

FIG. 7 is a perspective view of a component of the mattress assembly shown in FIG. 1.

DETAILED DESCRIPTION

The exemplary embodiments are discussed in terms of mattresses, such as, for example, mattress assemblies that include a dual-side inner core having two different comfort choices. The present disclosure may be understood more readily by reference to the following detailed description of the disclosure. It is to be understood that this disclosure is not limited to the specific devices, methods, conditions or parameters described and/or shown herein, and that the terminology used herein is for the purpose of describing particular embodiments by way of example only and is not intended to be limiting of the claimed disclosure.

Also, as used in the specification and including the appended claims, the singular forms “a,” “an,” and “the” include the plural, and reference to a particular numerical value includes at least that particular value, unless the context clearly dictates otherwise. Ranges may be expressed herein as from “about” or “approximately” one particular

value and/or to “about” or “approximately” another particular value. When such a range is expressed, another embodiment includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another embodiment. It is also understood that all spatial references, such as, for example, horizontal, vertical, top, upper, lower, bottom, left and right, are for illustrative purposes only and can be varied within the scope of the disclosure. For example, the references “upper” and “lower” or “top” and “bottom” are relative and used only in the context to the other, and are not necessarily “superior” and “inferior”.

The following discussion includes a description of a mattress assembly 10. Alternate embodiments are also disclosed. Reference will now be made in detail to the exemplary embodiments of the present disclosure.

The components of mattress assembly 10 can be fabricated from materials including metals, synthetic polymers, ceramics and/or their composites, depending on the particular application and/or preference. For example, the components of mattress assembly 10, individually or collectively, can be fabricated from materials such as stainless steel alloys, aluminum, commercially pure titanium, titanium alloys, Grade 5 titanium, super-elastic titanium alloys, cobalt-chrome alloys, stainless steel alloys, superelastic metallic alloys (e.g., Nitinol, super elasto-plastic metals, such as GUM METAL® manufactured by Toyota Material Incorporated of Japan), ceramics, thermoplastics such as polyaryletherketone (PAEK) including polyetheretherketone (PEEK), polyetherketoneketone (PEKK) and polyetherketone (PEK), carbon-PEEK composites, PEEK-BaSO₄ polymeric rubbers, polyethylene terephthalate (PET), fabric, silicone, polyurethane, silicone-polyurethane copolymers, polymeric rubbers, polyolefin rubbers, hydrogels, semi-rigid and rigid materials, elastomers, rubbers, thermoplastic elastomers, thermoset elastomers, elastomeric composites, rigid polymers including polyphenylene, polyamide, polyimide, polyetherimide, polyethylene, and epoxy. Mattress assembly 10 may be made from materials such as, for example, one or more of cotton, kapok, flax, linen, jute, ramie, hemp, kenaf, bamboo, coir, sisal, silk, wool (sheep), alpaca, llama, goat (mohair, cashmere), rabbit (angora), camel, horse, yak, vicuna, qiviut, guanaco, rayon, azlon, lyocell, acetate, triacetate, rubber, polyester, olefin, nylon, acrylic, modacrylic, aramid, spandex. Mattress assembly 10 is made such that mattress assembly 10 is compliant with all U.S. Federal Mattress Regulations, such as, for example, 16 C.F.R. §§ 1632 and 1633. Various components of mattress assembly 10 may have material composites, including the above materials, to achieve various desired characteristics such as strength, rigidity, elasticity, compliance, mechanical performance, and durability. The components of mattress assembly 10, individually or collectively, may also be fabricated from a heterogeneous material such as a combination of two or more of the above-described materials. The components of mattress assembly 10 may be monolithically formed, integrally connected or include fastening elements and/or instruments, as described herein.

Mattress assembly 10 includes a cover 12 that defines a pocket 14. In some embodiments, cover 12 includes only one pocket, such as, for example, pocket 14. In some embodiments, cover 12 includes a plurality of discrete pockets. In some embodiments, cover 12 includes a fastener, such as, for example, a zipper 16 to provide access to pocket 14. In some embodiments, cover 12 comprises a porous material, such as, for example, a porous fabric that allows air

to move in and out of pocket through cover 12. In some embodiments, cover 12 comprises a breathable material, such as, for example, a breathable fabric that allows air to move in and out of pocket through cover 12. In some embodiments, cover 12 is made entirely of a porous material and/or a breathable material. That is, no portion of cover 12 is waterproof. In some embodiments, cover 12 is washable. That is, cover 12 is configured to be washed in a standard washing machine.

One or a plurality of spring packs 18 is/are positioned within pocket 14. Spring pack 18 comprises a plurality of springs 20 positioned within a pouch 22. Springs 20 are enclosed within pouch 22. Spring pack 18 comprises one or a plurality of rows of springs 20 and one or a plurality of columns of springs 20. In some embodiments, spring pack 18 includes a plurality of rows of springs 20 and a plurality of columns of springs 20. In some embodiments, spring pack 18 includes a plurality of strings of springs, as described in U.S. Patent Application No. 62/347,199 to the inventor of the current application, which is incorporated by reference herein, in its entirety. In some embodiments, springs 20 are each positioned within a pocket, such as, for example, a fabric pocket. The pockets may be coupled to one another to form a string of pockets that each include one of springs 20 therein. In some embodiments, the string of pockets includes one or more slits between adjacent pockets to allow springs 20 to move independently of one another. In some embodiments, the string of pockets includes one or more slits that extend through a top surface of the string of pockets between adjacent pockets and/or one or more slits that extend through a bottom surface of the string of pockets between adjacent pockets. In some embodiments, spring pack 18 is permanently fixed to cover 12. In some embodiments, spring pack 18 is removably disposed within pocket 14.

Pouch 22 includes a bottom surface that directly engages cover 12 and an opposite top surface 24. Springs 20 are positioned between the bottom surface of pouch 22 and surface 24. In some embodiments, surface 24 comprises a waterproof material such that surface 24 defines a waterproof barrier of pouch 22 that prevents liquids from passing through surface 24 to prevent liquids or moisture from entering pouch 22 through surface 24. In some embodiments, surface 24 is entirely waterproof. That is, the entire area of surface 24 that is defined by the perimeter of surface 24 is waterproof. In some embodiments, surface 24 is non-porous. In some embodiments, surface 24 comprises a waterproof fabric. In some embodiments, surface 24 comprises a waterproof polymer. In some embodiments, surface 24 comprises a thermoplastic polyurethane. In some embodiments, surface 24 comprises a thermoplastic polyurethane breathable waterproof fabric. In some embodiments, surface 24 comprises a porous fabric that is coated with a waterproof material.

In some embodiments, mattress assembly 10 does not include any springs, such as, for example, spring pack 18. In some embodiments, mattress assembly 10 includes a material, such as, for example, a foam block in place of spring pack 18.

An inner core 26 is removably positioned within pocket 14 such that core 26 directly engages surface 24 of spring pack 18. Core 26 includes opposite first and second portions 28, 30 and a middle portion 32 between portions 28, 30. Portion 28 has a firmness that is different than a firmness of portion 30 to provide two different comfort choices. For example, core 26 can include a firmer side for toddlers and a softer side for tweens. In some embodiments, portion 32 has a firmness that is less than a firmness of portion 28 and

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a firmness of portion 30. Core 26 can be flipped as the child grows to prevent the need to buy another mattress as the child ages. For example, the same mattress can be used from when the child is 2 or 3 years old until the child is 12 or 13 years old. Core 26 is configured to be moved between a first configuration in which portion 28 directly engages surface 24 of spring pack 18 and a second configuration in which portion 30 directly engages surface 24 to alter a firmness of a sleep surface 34 of mattress assembly 10.

In some embodiments, core 26 comprises a matrix of fibers. The fibers that make up portion 28 are more densely packed than the fibers that make up portion 30 such that portion 28 is firmer than portion 30. In some embodiments, the fibers are made from a polymer, such as, for example, polyurethane. In some embodiments, core 26 consists of the fibers. That is, core 26 does not include any material or components other than the fibers. In some embodiments, core 26 comprises a mat of electrospun fibers. A first side of core 26 that defines portion 28 and a second side of core 26 that defines portion 30 are heat pressed to provide portions 28, 30 with selected firmnesses. In some embodiments, portion 28 is heat pressed for a longer duration of time and/or at a higher temperature than portion 30 to make portion 28 firmer than portion 30. In some embodiments, core 26 is washable. That is, core 26 is configured to be washed using water to remove any urine or other material on the fibers that make up core 26. In some embodiments, core 26 is free of any foam.

In operation and use, core 26 may be positioned within pocket 14 such that portion 30 directly engages surface 24 of spring pack 18 and portion 28 is positioned adjacent to sleep surface 34 of mattress assembly 10 to provide sleep surface 34 with a firmness sufficient to accommodate infants or toddlers. If the infant or toddler should wet the bed, the urine will move through cover 12 and core 26 and onto surface 24 of spring pack 18. To clean mattress assembly 10, zipper 16 of cover 12 can be moved from a closed configuration to an open configuration. Spring pack 18 and core 26 may be removed from pocket 14 when zipper 16 is in the open configuration. Cover 12 may be cleaned in a standard washing machine, for example. Spring pack 18 may be cleaned by wiping any urine off of surface 24. A cleaning material, such as, for example, bleach or vinegar may be applied to surface 24 before surface 24 is wiped clean. Core 26 may be cleaned by rinsing core 26 with water, for example. After cover 12, spring pack 18 and core 26 are cleaned, spring pack 18 may be repositioned within pocket 14. Core 26 is then inserted into pocket 14 such that portion 30 directly engages surface 24 of spring pack 18 and portion 28 is positioned adjacent to sleep surface 34 of mattress assembly 10. Zipper 16 is then moved from the open configuration to the closed configuration.

As the infant or toddler ages, he or she may desire that sleep surface 34 be less firm. To alter the firmness of sleep surface 34, zipper 16 of cover 12 can be moved from the closed configuration to the open configuration. Core 26 is removed from pocket 14 when zipper 16 is in the open configuration. Core 26 is flipped and is repositioned within pocket such that portion 28 directly engages surface 24 of spring pack 18 and portion 30 is positioned adjacent to sleep surface 34 of mattress assembly 10 to provide sleep surface 34 with a firmness sufficient to accommodate older children. In some embodiments, spring pack 18 remains within pocket 14 while core 26 is being flipped. That is, cover 12 and spring pack 18 are not flipped to alter the firmness of sleep surface 34. Rather, only core 26 is flipped, while cover 12 and spring pack 18 remain stationary. Should mattress

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assembly 10 become soiled, spring pack 18 and core 26 may be removed from pocket 14 and cover 12, spring pack 18 and/or core 26 may be cleaned, as discussed herein. After cover 12, spring pack 18 and/or core 26 is/are cleaned, spring pack 18 and core 26 may be repositioned within pocket 14, as discussed herein.

It will be understood that various modifications may be made to the embodiments disclosed herein. For example, features of any one embodiment can be combined with features of any other embodiment. Therefore, the above description should not be construed as limiting, but merely as exemplification of the various embodiments. Those skilled in the art will envision other modifications within the scope and spirit of the claims appended hereto.

What is claimed is:

1. A mattress comprising:

a cover defining a pocket and comprising opposite top and bottom walls, an outer surface of the top wall defining a sleep surface, the cover being made from a breathable material to allow air to move in and out of the pocket through the top and bottom walls; and

an inner core removably positioned within the cover, the inner core comprising opposite first and second portions, the first portion having a firmness that is greater than a firmness of the second portion,

wherein the inner core is configured to be moved between a first configuration in which the first portion directly engages the top wall and a second configuration in which the second portion directly engages the top wall to alter a firmness of the sleep surface,

wherein the second portion directly engages the bottom wall when the inner core is in the first configuration and the first portion directly engages the bottom wall when the inner core is in the second configuration, and

wherein the mattress further comprises a waterproof barrier positioned between the cover and the inner core such that the second portion of the inner core directly engages the waterproof barrier when the inner core is in the first configuration and the first portion directly engages the waterproof barrier when the inner core is in the second configuration.

2. The mattress recited in claim 1, further comprising a support positioned between the waterproof barrier and the bottom surface of the cover.

3. The mattress recited in claim 1, wherein the inner core comprises a matrix of fibers, the fibers that make up the first portion being more densely packed than the fibers that make up the second portion.

4. The mattress recited in claim 1, wherein the wherein the inner core comprises a matrix of polyurethane fibers, the fibers that make up the first portion being more densely packed than the fibers that make up the second portion.

5. The mattress recited in claim 1, wherein the inner core comprises a mat of electrospun fibers.

6. The mattress recited in claim 1, wherein the inner core comprises a mat of electrospun polymer fibers.

7. The mattress recited in claim 1, wherein the waterproof barrier is coated with a waterproof material.

8. The mattress recited in claim 1, wherein the waterproof barrier comprises a thermoplastic polyurethane.

9. The mattress recited in claim 1, wherein the cover includes a fastener to allow the inner core to be removed from the cover.

10. The mattress recited in claim 1, wherein the cover includes a zipper to allow the inner core to be removed from the cover.

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11. The mattress recited in claim 1, wherein the mattress is free of springs.

12. The mattress recited in claim 2, wherein the support is springs or foam material.

13. The mattress recited in claim 1, wherein the inner core comprises a foam.

14. The mattress recited in claim 1, wherein the cover includes a top section and a bottom section, the bottom section being coupled to the top section by a fastener.

15. The mattress recited in claim 1, wherein the cover includes a top section and a bottom section, the bottom section being coupled to the top section by a zipper.

16. A mattress comprising:

a cover defining a pocket and comprising opposite top and bottom walls, an outer surface of the top wall defining a sleep surface, the cover being made from a breathable material to allow air to move in and out of the pocket through the top and bottom walls;

an inner core removably positioned within the cover, the inner core comprising a mat of fibers having opposite first and second portions, the first portion having a firmness that is greater than a firmness of the second portion; and

a waterproof barrier positioned between the cover and the inner core;

wherein the inner core is configured to be moved between a first configuration in which the first portion directly engages the top wall and a second configuration in which the second portion directly engages the top wall to alter a firmness of the sleep surface,

wherein the second portion directly engages the waterproof barrier when the inner core is in the first configuration and the first portion directly engages the waterproof barrier when the inner core is in the second configuration.

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17. A mattress consisting of:

a cover defining a pocket and comprising opposite top and bottom walls, an outer surface of the top wall defining a sleep surface, the cover being made from a breathable material to allow air to move in and out of the pocket through the top and bottom walls;

an inner core removably positioned within the cover, the inner core comprising a mat of fibers, the mat defining opposite first and second portions, the first portion having a firmness that is greater than a firmness of the second portion; and

a support positioned between the inner core and the bottom wall of the cover comprising opposite first and second surfaces, the first surface defining a waterproof barrier,

wherein the inner core is configured to be moved between a first configuration in which the first portion directly engages the top wall of the cover and a second configuration in which the second portion directly engages the top wall of the cover to alter a firmness of the sleep surface, and

wherein the second portion directly engages the first surface of the support when the inner core is in the first configuration and the first portion directly engages the first surface of the support when the inner core is in the second configuration.

18. The mattress recited in claim 17, wherein the first surface of the support defining a waterproof barrier prevents liquids from passing through said surface.

19. The mattress recited in claim 17, wherein the support comprises a spring pack.

20. The mattress recited in claim 17, wherein the support comprises a foam mass.

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